Application Program Development

Ch 02: Input, Processing and Output

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Reading Input from the Keyboard

- Most programs need to read input from the user
- Built-in `input()` functions reads input from standard input (keyboard)
  - Returns the data as a `string`
  - Does not automatically display a space after the prompt

```
>>> name = input('What is your name? ')
What is your name? Alan Turing
>>> print(name)
Alan Turing
```

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Reading Numbers with the input() Function

- `input()` funciton always returns a string
- Built-in functions convert between data types
    - `int(item)` converts *item* to an `int`
    - `float(item)` converts *item* to a `float`
    - **Nested function call**: are allowed, output from inner function becomes parameter input to outer function.
    - Type conversion only works if item is a valid numeric literal, otherwise throws exception

```
>>> hours = input('How many hours did you work? ')
How many hours did you work? 22
>>> type(hours)
<class 'str'>
>>> hours = int(hours)
>>> type(hours)
<class 'int'>
>>> print(hours)
22
>>> hours = int(input('How many hours did you work? '))
How many hours did you work? 33
>>> type(hours)
<class 'int'>
```

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Performing Calculations

- Math expression: perform calculation and gives a value
  - **Math operator**: tool for performing calculation
  - **Operands**: values operations are applied to
    - Variables can be used as operands
  - Resulting value typically assigned to a variable
- **Note**: Two types of division:
  - / operator performs floating point division
  - // operator performs integer division

Symbol	Operation	Description
+	Addition	Adds two numbers
-	Subtraction	Subtracts one number from another
*	Multiplication	Multiplies one number by another
/	Division	Divides one number by another and gives the result as a floating-point number
//	Integer division	Divides one number by another and gives the result as a whole number
%	Remainder	Divides one number by another and gives the remainder
**	Exponent	Raises a number to a power

Operator Precedence and Grouping with Parentheses

Python Operator Precedence

1. Operations enclosed in parentheses
   - Forces operations to be performed before others
2. Exponentiation `**`
3. Multiplication `*`, division `/` and `//`, and remainder `%`
4. Addition `+` and subtraction `-`

- Higher precedence performed first
- Same precedence operators execute from left to right

Converting Math Formulas to Programming Statements

- Operators required for any mathematical operation
- When converting mathematical expression to programming statement:
  - May need to add multiplication operators
  - May need to insert parentheses

Algebraic Expression	Python Statement
$y = 3\frac{x}{2}$	`y = 3 * x / 2`
$z = 3bc + 4$	`z = 3 * b * c + 4`
$a = \frac{x+2}{b-1}$	`a = (x + 2) / (b - 1)`
$y = 3x^2 + 4x - 5$	`y = 3 * x**2 + 4 * x - 5`

Mixed-Type Expressions and Data Type Conversion

Data type resulting from math operation depends on **data types** of operands

- Two `int` values: result is an `int`
- Two `float` values: result is a `float`
- `int` and `float`: `int` converted to `float`, result of operation is `float`
- Type conversion of `float` to `int` causes truncation of fractional part

Breaking Long Statements into Multiple Lines

- Long statements cannot be viewed on screen without scrolling and cannot be printed without cutting off (see PEP8 Python Formatting Standards)
- Multiline continuation character \: Allows to break a statement into multiple lines (PEP8 recommends breaking before operator for readability)

```
result =    var1 * 2 + var2 * 3 \
          + var3 * 4 + var4 * 5
```

- Any part of a statement that is enclosed in parentheses can be broken without the line continuation character

```
result =  (  var1 * 2 + var2 * 3
           + var3 * 4 + var4 * 5)


print("Monday's sales are", monday,
      "and Tuesday's sales are", tuesday,
      "and Wednesday's sales are", wednesday)
```

String Concatenation and Repetition

You can use string concatenation to combine strings

```
my_str = 'Hello' + 'world'
```

The * operator performs string repetition

```
>>> my_str = 'Hello ' + 'World '
>>> print(my_str)
Hello World
>>> big_hello = my_str * 3
>>> print(big_hello)
Hello World Hello World Hello World
```

More About the `print()` Function

- `print()` function displays line of output
  - Newline character at end of printed data
  - Special argument `end='delimiter'` causes `print()` to place *delimiter* at end of data instead of newline character
- `print()` funciton uses space as item separator
  - Special argument `sep='delimiter'` causes `print()` to use *delimiter* as item separator
- Special characters appearing in string literal (see Escape Sequences in C)
  - Preceded by backslash \
    - Examples: newline `\n` horizontal tab `\t`
  - Treated as commands embedded in string

Displaying Formatted Output with F-strings (1 of N)

- An **f-string** is a special type of string literal in Python that is prefixed with the letter `f`
- **F-strings** support placeholders for variables

```
>>> name = 'John VonNeumann'
>>> print(f'Hello {name}!')
Hello John VonNeumann!
```

- Placeholders can be expressions that are evaluated

```
>>> print(f'The value is {10 + 2}.')
The value is 12.
>>> val = 10
>>> print(f'The value is {val + 2}.')
The value is 12.
```

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Displaying Formatted Output with F-strings (2 of N)

- Format specifiers can be used with placeholders

```
>>> print(f'{num:.2f}')
123.46
```

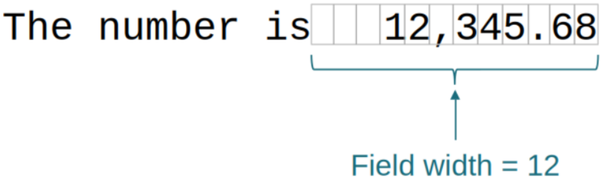
- `.2f` means:
  - round the value to 2 decimal places
  - display the values as a floating-point number

See PEP 3101 – Advanced String Formatting for a full list of all format specifiers

Displaying Formatted Output with F-strings (3 of N)

Specifying a minimum field width:

```
>>> num = 12345.6789
>>> print(f'The number is {num:12,.2f}')
The number is    12,345.68
```



Displaying Formatted Output with F-strings (4 of N)

- Aligning values within a field
    - use < for left alignment
    - Use > for right alignment
    - Use ^ for center alignment

```
>>> name = 'Alan Turing'
>>> title = 'Computer Scientist'
>>> salary = 4567.1234
>>> print(f'{name:a<20s}|{title:b^40s}|{salary:c>20.2f}')
Alan Turingaaaaaaaaa|bbbbbbbbbbbComputer Scientistbbbbbbbbbbb|ccccccccccccc4567.12
```

Displaying Formatted Output with F-strings (5 of N)

- The order of designators in a format specifier
    - When using multiple designators in a format specifier, write them in this order:

```
[alignment][width][,][.precision][type]
```

- Example:

```python
print(f'{number:^10,.2f}')
```

Magic Numbers

- A magic numbers is an unexplained numeric value that appears in a program's code.

```
amount = balance * 0.069
```

- What is the value 0.069? An interest rate? A fee percentage? Only person who wrote the code knows for sure (bad communication).
  - Can be difficult to determine its purpose
  - If it needs to be used in multiple places, can lead to bugs if needs to be changed.
  - You might make a mistake each time you retype in the magic number.

Named Constants

- **Use named constants instead of magic numbers**
- A named constant is a name that represents a value that does not change during the program's execution.

```python
INTEREST_RATE = 0.069
```

- Python PEP8 style states named constants should use `UPPER_CASE_SNAKE_VARIABLES`

Advantages of Using Named Constants

- Named constants make code self-explanatory, self-documenting, readable code
- Named constants make code easier to maintain
- Named constants help prevent typographical errors and bugs

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References